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Geotectonics of the cratonic margin from paleomagnetism of the Middle Proterozoic Aldridge (Prichard) Formation and Moyie sills of British Columbia and Montana

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ABSTRACT

The Precambrian Aldridge (Prichard) Formation crops out at or near the base of the Middle Proterozoic Belt-Purcell Supergroup in the Purcell anticlinorium, and the Moyie sills intrude the Lower and Middle Aldridge Formation. Paleomagnetic analysis was done on specimens from 27 sites in layered argillites of the Aldridge and from 30 sites in the Moyie metagabbros using AF and thermal (Th) demagnetization methods. Fold and contact tests show that no primary remanence components were isolated. However, three post-folding metamorphic components were found: A was found in 9 argillite (AF, 22.9°, 64.1°, α_{95} = 9.8°) and 11 sill sites (AF, 35.2°, 62.0°, α_{95} = 7.1°); B was found in 8 argillite (AF, 320.4°, 52.0°, α_{95} = 11.8°) and 10 sill sites (Th, 303.6°, 56.9°, α_{95} = 7.8°); and C was found in 19 argillite sites (AF, 209.0°, 3.1°, α_{95} = 6.0°). All three components include normal and reversed sites. The A, B, and C poles are discordant with the positions expected for known metamorphic or intrusive events, but correction for a simple 37° clockwise rotation leads to concordancy of the A pole with the ≈175 Ma position, B with the ≈810 Ma position, and C with the ≈1400 Ma position on the APWP for the North American craton. These ages coincide with the known Laramide, Goat River, and East Kootenay orogenies. Evidently, the terrane housing the Purcell Anticlinorium: *(1)* has always been part of the North American craton with little or no translation; *(2)* was folded and metamorphosed in the East Kootenay orogeny prior to 1400 Ma; *(3)* was thermally overprinted in the Goat River orogeny; and *(4)* was thermally overprinted and then rotated ≈37°±12° clockwise as Terrane I of the Cordillera impacted from the southwest against the craton during the mid-Mesozoic Laramide orogeny.

INTRODUCTION

The Middle Proterozoic Belt-Purcell Supergroup is exposed in southeastern British Columbia and in contiguous parts of Alberta, Montana, Idaho, and Washington (**Figure 1**). The Aldridge Formation in Canada and the equivalent Prichard Formation in the United States are stratigraphically near the base of the supergroup. The Moyie sills intrude the lower part of the Aldridge (Prichard) Formation. The study area is in the Purcell anticlinorium where paleomagnetic fold tests can indicate if a given remanence component was formed before or after folding. The object of the study was to determine if this Precambrian terrane was rotated or translated relative to the North American craton, as was found in Mesozoic rocks of Terrane I of the Cordillera to the immediate west (Symons and Litallien, 1984).

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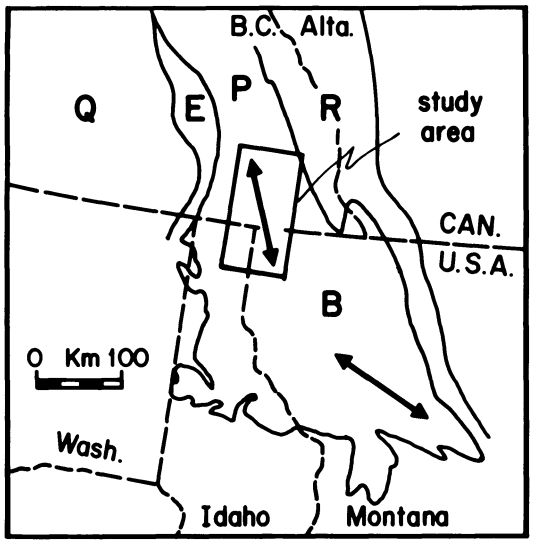


Figure 1. Location of the study area in the Belt-Purcell Supergroup (B, P) outcrop area. The arrows denote the structural trends. The boundaries of the Rocky Mountain Belt (R) and the Quesnellia (Q) and Eastern (E) subterranes of Terrane I are also shown.

Geology

The Belt-Purcell Supergroup rests unconformably along its eastern margin on a plutonic-metamorphic basement that yields radiometric ages of $\approx 1700 \pm 100$ Ma (Burwash and others, 1962; Giletti, 1968) and probably of ≈ 1540 Ma (Reid and others, 1973; Armstrong, 1975; Evans, 1986). This sets a maximum age for the supergroup. In Canada the Aldridge is comprised of ≈ 5000 m of wackes and argillites in a flysch sequence that overlies ≈ 2000 m of argillites and quartzites of the Fort Steele Formation at the base of the Purcell Supergroup (Hamilton and others, 1982). The stratigraphic equivalent of the Aldridge in Montana is the 7500-m-thick Prichard Formation at the base of the Belt Supergroup (Cressman, 1984). The Aldridge (Prichard) is overlain in turn by several kilometers of wackes, argillites, and quartzites with minor coarse clastic, carbonate, and basalt units in the supergroup, and above these by younger Proterozoic and Paleozoic formations.

The Moyie sills are metagabbroic in composition and up to 350 m thick. They are found in only the Aldridge (Prichard) Formation. They show minor crosscutting relationships in the Lower Aldridge division but are essentially stratabound in the Middle Aldridge division. One sill, the Crossport C sill near Eastport, Idaho, gives a U-Pb zircon age of 1433 ± 10 Ma which dates intrusion, as well as Rb-Sr and K-Ar ages of 1285 ± 165 and 850 ± 45 Ma, respectively, which reflect subsequent metamorphism (Zartmann and others, 1982). Altered wall rock from the Sullivan Pb-Zn-Ag deposit in the Lower Aldridge near Kimberly, B.C., gives a K-Ar biotite age of 1436 Ma (LeCouteur, 1979). Thus, 1435 Ma is likely both the actual age of the sills and the minimum age for the Aldridge Formation. This result is consistent with the U-Pb zircon age of 1370 ± 110 Ma from porphyritic granite plutons in the Salmon-Leesburg area of Idaho and the Rb-Sr isochron age of 1305 ± 152 Ma found for the Hellroaring Creek granodiorite stock, because these felsic plutons cut Prichard sediments and Moyie sills and post-date much of the deformation and regional metamorphism in the Purcell anticlinorium (Ryan and Blenkinsop, 1971, recalculated using $\lambda = 1.42 \times 10^{-11}$ year^{-1}; Evans, 1986). From the metamorphic mineralogy and isotope data, McMechan and Price (1982) concluded that regional metamorphism reached lower middle greenschist facies temperatures of about 375°C in the Belt-Purcell rocks of the study area. They also identified three metamorphic episodes: *(1)* the East Kootenay orogeny from 1350 to 1300 Ma, which produced most of the folding; *(2)* the Goat River orogeny from 900 to 800 Ma; and *(3)* the Laramide orogeny from Jurassic to early Tertiary time. The Laramide orogeny signals the accretion of allochthonous Terrane I onto the craton in the Early Jurassic (Monger and others, 1982; Brown and others, 1986). Mesozoic rocks in the Quesnellia subterrane of Terrane I to the immediate west of the study area were rotated clockwise $\approx 43°$ and translated northward $\approx 13°$ relative to the craton (Symons and Litalien, 1984). Cressman (1984) reported that sole marks on turbidites in the Aldridge (Prichard) Formation to the northwest have been rotated $\approx 45°$ clockwise on Laramide thrust faults relative to transport directions found to the southeast. Archibald and others (1983, 1984) and Price (1986) consider that rocks of the Purcell anticlinorium were deeply buried in Jurassic time, were rapidly uplifted in Early Cretaceous time and overthrust into a supracrustal position, and were then intruded by Late Cretaceous to early Tertiary granitic intrusions. Lithoprobe seismic reflection data indicate that a series of W-dipping thrust faults is present in the supracrustal